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Abir Hasan

SKILLS

Languages: C/C++, Java, JavaScript, TypeScript, PHP, Python, MySQL, HTML, CSS, YAML, LaTeX

Frameworks & Libraries: React (Vite), Node.js, Express.js, MongoDB, Flask, Expo/React Native, Tailwind CSS, Bootstrap

Backend & APIs: REST APIs, WebSockets (PHP Ratchet), gRPC, JWT-based Auth, PHPMailer

AI/ML & Data: XGBoost, PyTorch, NumPy, Pandas, Model Serialization (Pickle), Jupyter Notebooks

Infrastructure & Tooling: Nginx, CI/CD (GitHub Actions/YAML), Git, Linux (Ubuntu), Vite, ESLint

EXPERIENCE

Contract Software Engineer - Qaders Heaven & Student Study Abroad Agency System

November 2025 - Present

- Contracted to deliver and maintain two production systems for the same client: a live property-management platform (Qaders Heaven) and a student study-abroad agency system.
- Owned end-to-end delivery: React/Vite frontends, PHP/MySQL/JWT APIs, and nginx-backed production deployments; responsible for performance tuning, analytics queries, and hotfix rollouts.
- Implemented branch-aware CI/CD: GitHub-driven YAML workflows, and push-to-deploy server-side builds that produce artifacts directly in the server directory.
- Delivered handover documentation and continue to provide iterative updates, bug fixes, and deployment support.

PROECTS

Qaders Heaven - Property management web app (Production, Contract)

Summary: Production-level property management system (hosted on a live server and used by real users) covering properties, tenants, billing, reporting and an AI chat helper.

- Implemented a modular React + Vite SPA with Tailwind CSS and role-protected routes for Admin/Tenant flows.
- Built report generation UI and client-side PDF export using jsPDF, including paginated PDF output and summary metrics (total billed, collected, outstanding, collection rate).
- Developed PHP REST endpoints for bill generation and monthly billing workflows.
- Created analytics/home-summary SQL queries and an AI-chat backend hook that executes KPI queries and returns structured responses.
- Configured nginx for serving the app and reverse proxying backend APIs in production.
- Implemented CI/CD to support parallel development and production using environment variables (.env) and branch-based workflows; connected server to GitHub and automated builds/deploys via YAML scripts triggered on git push, enabling direct server-side builds.

Source: https://github.com/abirinpajamas/Qders-Heaven

ADHD-prediction-model and Deployment — Training, experimentation & model artifact (Jupyter)

Summary: Reproducible notebooks that ingest and merge multi-source clinical and sensor datasets, perform EDA and preprocessing, and produce a production-ready XGBoost classifier artifact consumed by the prediction API.

- Merged datasets and CPT outputs into a feature matrix (X) and binary target (y), dropped noisy/irrelevant columns and visualized class-conditional feature distributions for sanity checks.
- Applied mean imputation, StandardScaler normalization, and explicit column pruning to stabilize downstream training.
- Trained a RandomForest (n=100) as an importance estimator and used SelectFromModel to produce a reduced feature set (documented selected feature names and dimensionality reduction).
- Baseline modelling with XGBClassifier and stratified train_test_split.
- Performed a hyperparameter sweep via GridSearchCV with StratifiedKFold (n_splits=5) over n_estimators, learning_rate, max_depth, subsample, colsample_bytree, and gamma to find the best estimator.
- Evaluated using classification report, test-set accuracy and precision-recall AUC (plotted PR curve) to account for class imbalance sensitivity.
- Standardized the prediction contract as JSON and serialized the final model artifact to a pickle file for stateless inference.

- Implemented a Flask endpoint that accepts JSON feature arrays, converts to numpy arrays, and returns predictions as JSON.
- Deployed the pickled model on Render.

Tech: Python, pandas/numpy, scikit-learn (imputer, scaler, SelectFromModel, GridSearchCV), xgboost, matplotlib, pickle, Jupyter.
Source: https://github.com/abirinpajamas/adhd-prediction-ml-model

ADHD Mobile App — Screening & movement data collector

Summary: Expo app that collects accelerometer sessions and questionnaire responses, computes per-epoch features, and posts them to a prediction endpoint for screening.

- Implemented accelerometer collection pipeline (32 Hz sampling, 60s epochs, buffering, movement intensity features) with lifecycle safe start/stop and session backups.
- Built questionnaire flows that transform responses into features and call a remote XGBoost prediction Flask API; integrated FileSystem + Sharing for data export.
- Added developer helper scripts to reset project scaffolding.

Tech: Expo (React Native), TypeScript, expo-sensors, expo-file-system, REST integration.
Source: https://github.com/abirinpajamas/adhd-app

UniPlan — University planning & community portal

Summary: Campus utility platform with routine planner, faculty reviews, course scheduling, JWT-auth PHP APIs and WebSocket-driven comment/like features.

- Built React+TS frontend with modular components for course routines, CGPA calculator and faculty review.
- Implemented JWT-secured PHP endpoints for saving routines and review CRUD, including prepared-statement DB interactions and token decoding.
- Added a PHP WebSocket server to enable real-time updates for comments/likes and secure server-side review actions tied to JWT authorization.
- Integrated client-side token handling and route protection.
- Implemented a critical recursive backtracking routine-generation algorithm that constructs conflict-free course schedules while respecting credit targets, class day pairs(eg, STRA), theory-lab pairing constraints and faculty preferences. The algorithm:

Tech: React + TypeScript, Vite, Bootstrap, PHP (MySQL), JWT auth, WebSocket server.

Source: https://github.com/abirinpajamas/uniplan-app

EDUCATION

North South University — *Bsc in CSE*

September 2022 - Present

Relevant Coursework: Data Communication & Network, Software Engineering, Theory of Computation, Concept of Programming Language, Machine Learning